# Rocket Turbine Studio

AliAerospace

## 1 Overview

This document presents the input and output parameters used by the Rocket Turbine Studio program to develop and optimise axial-turbine rotor blades and vaned stators for turbopumps. The methodology employed is novel and is a proposed update developed by S. Sudhof [2] to the eleven parameter method developed by L. Pritchard for aerofoil shaping [1]. Understanding the parameters which control the curves for the rotor and stator is therefore vital, as some of the conventions are unique.

The program is intended to be used alongside CFD to further optimise the geometries of both rotor and stator. The workflow is outlined within the GitHub.

Please refer to [1] and [2] for further descriptions regarding the nomenclature.

# 2 Cycle inputs

| Code name | Symbol | Meaning | Unit |
|---|---|---|---|
| `cycle.P` | $P$ | Required shaft power | W |
| `cycle.mdot` | $\dot{m}$ | Turbine mass flow rate | kg/s |
| `cycle.d_m` | $d_m$ | Mean turbine diameter | mm |
| `cycle.omega` | $\omega$ | Rotational speed | rad/s |
| `cycle.reaction` | $R_m$ | Degree of reaction at the mean radius | – |
| `cycle.beta_1` | $\beta_1$ | Stator inlet metal angle | deg |
| `cycle.beta_2` | $\beta_2$ | Stator outlet metal angle | deg |
| `cycle.T_01` | $T_{01}$ | Stator inlet total temperature | K |
| `cycle.c_p` | $c_p$ | Specific heat capacity at constant pressure | J/(kg K) |
| `cycle.gamma` | $\gamma$ | Ratio of specific heats | – |
| `cycle.Mmol` | $M_{\mathrm{mol}}$ | Gas molar mass | kg/kmol |
| `cycle.p_01` | $p_{01}$ | Stator inlet total pressure | Pa |
| `cycle.degree_of_admission` | – | Fraction of flow allowed in the rotors circumferential arc | – |
| `cycle.delta_3` | $\delta_3$ | Rotor inlet flow deviation angle | deg |
| `cycle.gamma_ri` | $\gamma_{ri}$ | Rotor inlet wedge angle | deg |
| `cycle.eta_m` | $\eta_m$ | Mechanical efficiency | – |
| `cycle.eta_n` | $\eta_n$ | Nozzle efficiency | – |
| `cycle.n` | $n$ | Inlet whirl vortex exponent | – |
| `cycle.m` | $m$ | Work distribution vortex exponent | – |

# 3 Rotor inputs

| Code name | Symbol | Meaning | Unit |
|---|---|---|---|
| `rotorValues.chord` | $c_r$ | Rotor section chord | mm |
| `rotorValues.N` | $N_r$ | Number of rotor blades | – |
| `rotorValues.unguided_turning` | $\Gamma$ | Pritchard unguided turning angle | deg |
| `rotorValues.r_le_ratio` | $r_{le}/c_z$ | Ratio of leading edge radius to axial chord | – |
| `rotorValues.r_te_ratio` | $r_{te}/c_z$ | Ratio of trailing edge radius to axial chord | – |
| `rotorValues.t_ri` | $t_{ri}$ | Distance used to locate rotor point $P_3$ (at rotor inlet) | mm |
| `rotorValues.gamma_turning_ri` | $\Gamma_{ri}$ | Rotor inlet turning used to locate and orient $P_3$ | deg |
| `rotorValues.n_stagger` | $n$ | Rotor stagger curve exponent | – |

# 4 Generated Sudhof rotor variables

| Code name | Symbol | Meaning | Unit |
|---|---|---|---|
| `d_m` | $d_m$ | Diameter of the current rotor section | mm |
| `chord` | $c_r$ | Rotor chord | mm |
| `stagger_angle` | $\Phi$ | Stagger angle | deg |
| `gamma_turning_ri` | $\Gamma_{ri}$ | Inlet turning used to locate rotor point $P_3$ | deg |
| `gamma_ri` | $\gamma_{ri}$ | Rotor inlet wedge angle | deg |
| `beta_3` | $\beta_3$ | Rotor inlet metal angle | deg |
| `semi_minor_axis_ri` | $b_{ri}$ | Semi-minor axis of the rotor leading-edge ellipse | mm |
| `eccentricity_ri` | $e_{ri}$ | Eccentricity of the rotor leading-edge ellipse | – |
| `gamma_turning_ro` | $\Gamma_{ro}$ | Rotor outlet turning angle | deg |
| `gamma_ro` | $\gamma_{ro}$ | Rotor outlet wedge angle | deg |
| `beta_4` | $\beta_4$ | Rotor outlet metal angle | deg |
| `r_ro` | $r_{ro}$ | Rotor trailing-edge radius | mm |
| `t_ri` | $t_{ri}$ | Distance used to locate rotor point $P_3$ | mm |
| `curvature_P1` | $\kappa(P_1)$ | Curvature at rotor point $P_1$ | 1/mm |
| `curvature_P2` | $\kappa(P_2)$ | Curvature at rotor point $P_2$ | 1/mm |
| `quartic_pointedness` | $p$ | Pointedness of $P_1$–$P_2$ quartic | – |
| `quartic_bias` | $b$ | Bias of the $P_1$–$P_2$ quartic | – |
| `lambda_5` | $\lambda_5$ | First pressure side cubic tangent length | mm |
| `lambda_6` | $\lambda_6$ | Second pressure side cubic tangent length | mm |
| `s_bar` | $\bar{s}$ | First Paluszny curve parameter | – |
| `u_bar` | $\bar{u}$ | Second Paluszny curve parameter | – |
| `channel_expansion_ratio` | $t_{ri}/t_{ro}$ | Rotor channel expansion ratio | – |
| `N` | $N_r$ | Number of rotor blades | – |

# 5 Stator inputs

| Code name | Symbol | Meaning | Unit |
|---|---|---|---|
| `statorValues.r_te` | $r_{no}$ | Stator trailing edge radius (nozzle outlet) | mm |
| `statorValues.wedge_angle_1` | – | Stator inlet wedge angle | deg |
| `statorValues.gamma_no` | $\gamma_{no}$ | Stator outlet wedge angle | deg |
| `statorValues.semi_minor_axis_le` | $b$ | Semi-minor axis of the stator leading edge ellipse | mm |
| `statorValues.eccentricity_le` | $e$ | Eccentricity of the stator leading edge ellipse | – |
| `statorValues.w_45` | $w_{45}$ | Rational conic weight for stator curve $P_4$–$P_5$ | – |
| `statorValues.Rc` | $\bar{R}_c$ | Radius of curvature at stator point $P_2$ | mm |

| Code name | Symbol | Meaning | Unit |
|---|---|---|---|
| `statorValues.p_quartic` | $p$ | Pointedness of the stator $P_1$–$P_2$ quartic | – |
| `statorValues.b_quartic` | $b$ | Bias of the stator $P_1$–$P_2$ quartic | – |
| `statorValues.s_bar` | $\bar{s}$ | Paluszny parameter for stator curve $P_2$–$P_3$ | – |
| `statorValues.t_bar` | $\bar{t}$ | Paluszny parameter for stator curve $P_2$–$P_3$ | – |
| `statorValues.u_bar` | $\bar{u}$ | Paluszny parameter for stator curve $P_2$–$P_3$ | – |
| `statorValues.N` | $N_s$ | Number of stators | – |
| `statorValues.retraction` | – | Retraction used to locate throat point $P_{2p}$ | – |
| `statorValues.contraction_ratio` | $t/t_0$ | Throat to reference throat ratio | – |
| `statorValues.chord` | $c_s$ | Stator chord | mm |
| `statorValues.n_stagger` | – | Stator stagger equation curve exponent | – |
| `statorValues.admission` | – | Admission parameter | – |

# 6 Additional nomenclature

| Code name | Symbol | Meaning | Unit |
|---|---|---|---|
| – | $w$ | Relative velocity | m/s |
| – | $w_u$ | Tangential relative velocity | m/s |
| – | $c_u$ | Tangential absolute velocity | m/s |
| `cycle.beta_1` | $\beta_1$ | Stator inlet blade angle | deg |
| – | $d_s$ | Shroud diameter | mm |
| `cycle.d_m` | $d_m$ | Mean diameter | mm |
| – | $z$ | Axial coordinate | mm |

# 7 Defining Rotor and Stator Geometries

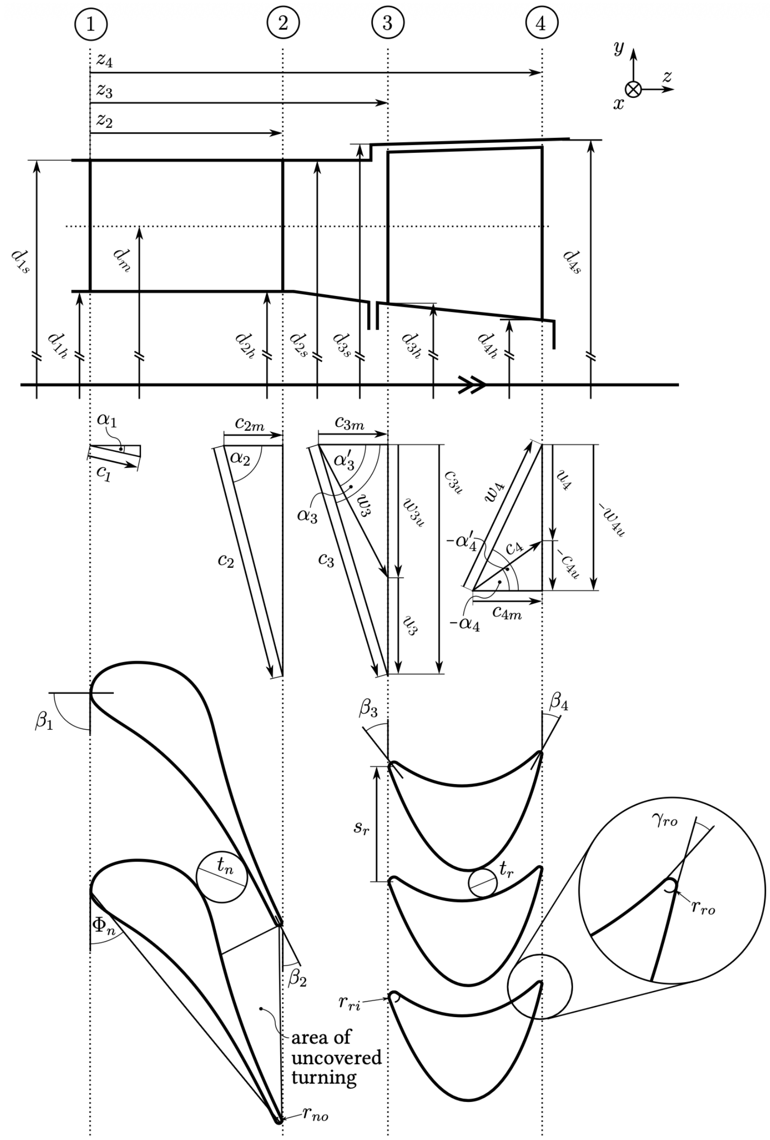


Figure 1: Annulus cross section with rotor and stator cross sections. Velocity triangles, pitch, throat, and turning parameters labelled [2].

## 7.1 Metal Angles

To determine the distribution of the metal angles along the span for both the stator and rotor, the inlet whirl vortex exponent $n$, work distribution vortex exponent $m$ are described in [5], where they are defined in Table 6.

Table 6: Vortex law exponents for various conditions. Values based on Aungier, Section 6.12 [5].

| Type | $n$ | $m$ | Description |
|---|---|---|---|
| Free vortex | 1 | 1 | Provides constant angular momentum, $rc_u$, and constant meridional velocity, $c_m$, from hub to shroud. |
| Constant swirl | 0 | 0 | Provides constant tangential velocity at rotor inlet, $c_{3u}$, across the span. |
| Exponential vortex | 0 | 1 | May allow constant-camber stator blades. However, is commonly associated with axial flow compressors. |
| Constant reaction | $-1$ | 1 | Only constant with $R = 1$. |
| Constant nozzle angle | $\cos^2 \alpha_2$ | $\cos^2 \alpha_2$ | Provides constant flow angle from hub to shroud for the nozzle exit. May allow for a constant section stator. |

## 7.2 Rotor

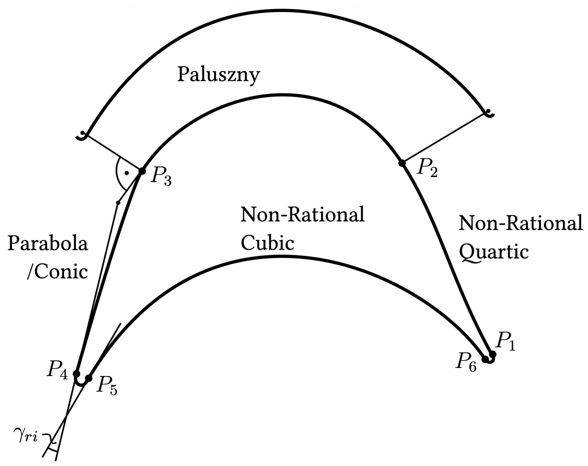


Figure 2: S. Sudhof's rotor parameterisation [2].

The workflow outlined in the GitHub

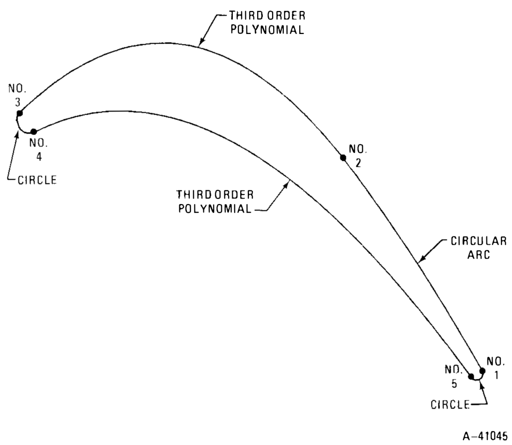


Figure 3: L. Pritchard's rotor parameterisation [1].

Please refer to [3] and [4] for descriptions of the Paluszny and Bezier curves as well as the other curves defined in Table 7.

Table 7: Sudhof rotor topology in `createSudhofRotor`.

| Segment or point | Definition |
|---|---|
| $P_1$–$P_2$ | Non-rational quartic Bezier |
| $P_2$–$P_3$ | Paluszny rational cubic |
| $P_3$–$P_4$ | Rational quadratic conic |
| $P_4$–$P_5$ | Elliptic arc |
| $P_5$–$P_6$ | Cubic Bezier |
| $P_6$–$P_1$ | Circular arc |

## 7.3 Stator

Please refer to [3] and [4] for descriptions of the Paluszny and Bezier curves as well as the other curves defined in Table 8.

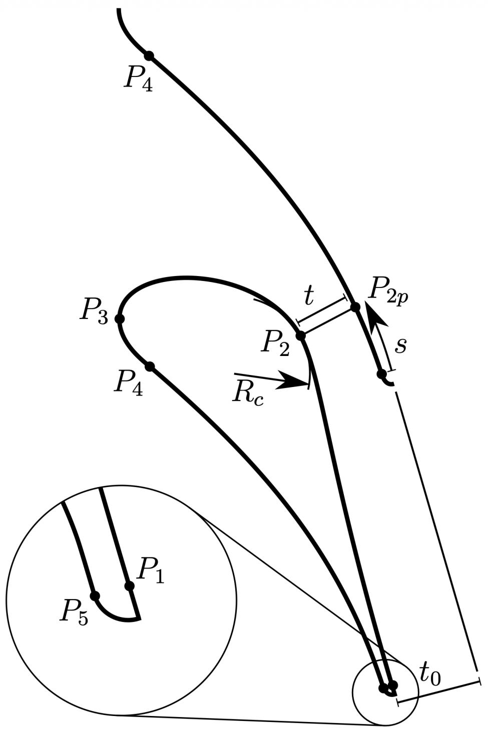


Figure 4: Stator parameterisation [2].

Table 8: Südhof stator topology in `createSudhofStator`.

| Segment or point | Definition | Use |
|---|---|---|
| $P_1$–$P_2$ | Non-rational quartic Bézier | – |
| $P_2$–$P_3$ | Paluszny rational cubic | – |
| $P_3$–$P_4$ | Elliptic arc | – |
| $P_4$–$P_5$ | Rational quadratic conic | – |
| $P_5$–tip | Quarter-circle portion | – |
| tip–$P_1$ | Straight segment | – |
| $P_{2p}$ | Throat contact point on the pressure side of neighbouring stator | Located along the neighbouring $P_4$–$P_5$ curve using the `statorValues.retraction` parameter. This defines one side of the minimum passage opening. |
| $t$ | Normal distance from $P_2$ to $P_{2p}$ | Stator throat distance. This is calculated from the reference opening $t_0$ using `statorValues.contraction_ratio`, where $t = (t/t_0)t_0$. |

# References


[1] L. Pritchard, “An Eleven Parameter Axial Turbine Airfoil Geometry Model,” *ASME 1985 International Gas Turbine Conference and Exhibit*, Paper 85-GT-219, 1985.

[2] S. Sudhof, *Development Techniques for Supersonic Turbines*, Dr.-Ing. dissertation, Institut f"ur Raumfahrtsysteme, Universit"at Stuttgart, 2020.

[3] M. Paluszny, F. Tovar, and R. R. Patterson, “$G^2$ Composite Cubic Bezier Curves,” *Journal of Computational and Applied Mathematics*, vol. 102, no. 1, pp. 49–71, 1999.

[4] L. Piegl and W. Tiller, *The NURBS Book*, 2nd ed., Springer, 1997.

[5] Ronald H. Aungier, *Turbine Aerodynamics: Axial-Flow and Radial-Inflow Turbine Design and Analysis*, ASME Press, New York, 2006. doi:10.1115/1.802418.